Finding and Expressing Patterns

For the expressions below use the "small n" problem solving strategy to make a guess for an function of n that should appear on the right hand side (RHS) of the expressions below, where the question mark is .

Expression 1

What is the sum of the first n odd integers or as a mathematical expression, what is $1+3+5+7+\cdots(2n-1) = ?$

For the "small n" strategy figure out the following:

n	sum	total
1	1	
2	1+3	
3	1+3+5	
4		
5		

Can you find a pattern? Can you express that idea mathematically as an expression involving n?

Repeat the same process as above for the next two expressions:

Expression 2

What is the sum of the first $n+1$ powers of 2 starting at 2^0, or $1 + 2 + 2^2 + \cdots + 2^n = ?$

Expression 3

What is the sum $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots \frac{1}{n(n+1)} = ?$ for $n \geq 1$, n an integer.

Expression 4

What is the sum of the first n positive integers, or $1 + 2 + 3 + 4 + \cdots n = ?$ for $n \geq 1$, n an integer. (Hint: Try finding a pattern for $2 + 4 + 6 + 8 + \cdots$ with a little help from expression 3 above.)